Statistical Foundations of Machine Learning

Practice Set 1

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1 Homework problems

Note: solutions to problems in this section are to be submitted and will be graded. Deadline for submission is Sun, 6th Sep.. Total marks : 15. Justify your answers and present all calculations/arguments clearly.

1. **(Counting and sampling)**

   (a) Let A be a 5×5 matrix with each entry being 0 or 1, selected equally likely and independently across entries. Find the probability that all the following conditions are satisfied: **[2 marks]**

   - the sum of the entries in the 2nd row is 4,
   - the sum of the entries in the 2nd column is 3, and
   - $A_{2,2} = 1$.

   (b) A subset is chosen uniformly at random from all possible subsets of $\{1, 2, \ldots, 10\}$. Find the probability that the product of the elements in the subset is odd. **[2 marks]**

2. **(Independence)**

   (a) Say A and B are two independent events in a probability space. Show that A and B^c (the complement event of B) are also independent. **[1 mark]**

   (b) Let

   $$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8\},$$

   and let $\mathcal{F} = 2^{\Omega}$. Define the probability measure $\mathbb{P}$ by

   $$\mathbb{P}(D) = \frac{|D|}{8}, \qquad \text{for every } D \subseteq \Omega.$$

   Define the events

   $$A = \{1, 2, 5, 6, 7, 8\},\ B = \{1, 3, 5, 6, 7, 8\},\ \text{and } C = \{1, 2, 3, 4\}.$$

   i. Are A and B conditionally independent given C? Justify your answer. **[2 marks]**

   ii. Are A and B independent? Justify your answer. **[2 marks]**

3. **(Bounds)** A fair six-sided die is rolled independently three times. Let

   $$\mathcal{E}_1 = \{\text{exactly one } 6 \text{ occurs among the three rolls}\}$$
   $$\mathcal{E}_2 = \{\text{at least one } 6 \text{ occurs among the three rolls}\}.$$

(a) Compute $\mathbb{P}(\mathcal{E}_1)$. **[1 mark]**

(b) Compute $\mathbb{P}(\mathcal{E}_2)$. **[1 mark]**

(c) Compute bounds on $\mathbb{P}(\mathcal{E}_2)$ using the first-order, second-order, and third-order Bonferroni inequalities. Compare the bounds with each other and the true value computed in the previous part. **[2 marks]**

4. A and B throw a pair of dice alternately. A wins the game if he gets a total of 9 and B wins the game if he gets a total of 4. If A starts the game, then find the probability that B wins. **[2 marks]**

2 Practice problems

Note: problems in this section are for practice, solutions not to be submitted

1. Give examples for the following cases. For each of them, explain the sample space and the events properly.
   Only assigning numerical values satisfying the conditions will not suffice

   (a) $P(A \cap B) > P(A)P(B)$ **[2 marks]**

   (b) $P(A \cap B) < P(A)P(B)$ **[2 marks]**

2. Alice and Bob are among a group of eight college students who are interested in debating. Three students are picked randomly from this group to represent the college at an intercollegiate fest. Let A be the event that Alice is among those picked, and B denote the event that Bob is among those picked.

   (a) $\mathbb{P}(A)$ **[1 mark]**

   (b) $\mathbb{P}(A \,|B)$ **[1 mark]**

   (c) $\mathbb{P}(A \cup B)$ **[1 mark]**

3. You flip a fair coin thrice and note the outcome (in $\{H, T\}^3$). Define the following events.

   A is the event that all tosses have the same result (all H or all T).

   B is the event that the you see at least one H and at least one T.

   C is the event that you see at most one H.

   (a) Are events A and B independent? **[1 mark]**

   (b) Are events B and C independent? **[1 mark]**

   (c) Are events A and C independent? **[1 mark]**

4. Show that if two events A and B are independent, then A^c and B^c are also independent. **[2 marks]**

5. Consider a knockout tournament with k rounds of matches including the final. Recall that in each round, half of the remaining players are 'knocked out' of the tournament. For example, the grand slam tennis tournaments start with 128 players and have seven rounds of matches. Define a suitable sample space for the random outcome of the tournament. Each element of this sample space need not be equally likely.

6. A course is taught by four instructors. Before every lecture, the instructors draw lots and one of them is randomly chosen to teach on that day. What is the probability that in N classes, all the lecturers would have taught at least once. Generalise to k instructors teaching the course.

7. Numbers 1, 2, 3, 4, 5, and 6 are randomly placed on a circle. What is the probability that they are placed in increasing order?

8. A and B play the following game of dice. Both roll a die once. A gets is assigned the number obtained on its roll. If B rolls a number other than one, it is assigned whatever it rolls. However, if B rolls a one, then it rolls again and will be assigned whatever it rolls the second time (even if it rolls a one). Whoever is assigned the highest value wins. If there is a tie A wins. What is the probability that A wins the game.

9. There are n urns and the r-th urn contains $(r-1)$ brown balls and $(n-r)$ black balls. You pick an urn at random and pick two balls at random without replacement. What is the probability that the second ball is black. What is the conditional probability that the second ball is black given that the first ball is black.

10. I choose a number uniformly in the range $[1, 1000000]$. Using inclusion exclusion principle, determine the probability that the chosen number is divisible by one or more of 4, 6, and 9.

11. An urn starts off with one black and one brown ball. Sample one ball from this urn and put the ball back into the urn along with another ball of the same color. Now there are three balls in the urn. Repeat this action N times, i.e., sample a ball and put the ball and another ball of the same color back urn into the urn. There are now $(N+2)$ balls in the urn. Find the probability that there k black balls in the urn.

12. Consider a best-of-five tennis match in which whoever wins three sets first wins the match. The players are evenly matched, i.e., each one wins a set with equal probability. Find the probability that loser has won k sets.

13. A coin is tossed independently n times. The probability of heads at each toss is p. At each time k $(k = 2, 3, ..., n)$, we get a reward if k^{th} toss was a head and the previous toss was a tail. Let A_k be the event that a reward is obtained at time k.

    (a) Are events A_k and A_{k+1} independent?

    (b) Are events A_k and A_{k+2} independent?

    Justify all answers.

14. There are three types of coins which have different probabilities of landing heads when tossed.

    - Type A coins are fair, with probability 0.5 of heads
    - Type B coins are biased and have probability 0.7 of heads
    - Type C coins are biased and have probability 0.9 of heads

    Suppose I have a drawer containing 5 coins: 2 of type A, 2 of type B, and 1 of type C. I reach into the drawer and pick a coin at random. Without showing you the coin I flip it once and get heads. What is the probability it is Type A? Type B? Type C?

15. Prove that

    (a) $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$

    (b) $\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) \geq \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots + \mathbb{P}(A_n) - (n-1)$

16. Consider two events A and B such that $\mathbb{P}(A) > 1 - \delta$, and $\mathbb{P}(B) > 1 - \delta$, for very small $\delta > 0$. Prove that $\mathbb{P}(A \cap B)$ is close to 1.

17. The Erdős-Rényi model $G(n, p)$ is a simple random graph model, where n is the number of nodes in the graph. In a social networking context, each node might represent a person. Every pair of nodes is connected by an edge with probability p. The occurrence of each edge in the graph is independent of other edges in the graph.

    Let B_n be the event that a graph randomly generated according to the $G(n, p)$ model has at least one isolated node. We want to show that $P(B_n) \leq n(1-p)^{n-1}$ and conclude that for any $\epsilon > 0$, if $p = p_n = (1+\epsilon)\frac{\ln(n)}{n}$, then

    $$\lim_{n \to \infty} P(B_n) = 0.$$

18. Each pack of Gold Cookies contains a coupon. There are n types of coupons available, and each is independent of previously obtained coupons. The probability of getting i^{th} coupon is p_i, with $\Sigma p_i = 1$. Let A_i denote the event that you have atleast one coupon of type i in the k already obtained coupons. Answer the following (assume $i \neq j$)-

    (a) $P(A_i)$

    (b) $P(A_i \cup A_j)$

    (c) $P(A_i | A_j)$

19. You are a contestant in a TV show. There are 3 doors leading to rooms of which 2 are empty and 1 contains a Mercedes. You are asked to choose one room and a correct guess will win you the Mercedes. After you select a room, the game master opens one of the door of another room that you did not select and also that do not contain the Mercedes. Prove that you will improve your chance to win the Mercedes if you change your choice from the initial guess to the remaining closed door. Mathematically

    – $\mathcal{E}_1$: Event of winning the Mercedes with the initial guess

- $\mathcal{E}_2$: Event of winning the Mercedes if he changes his initial guess
- $\mathcal{C}$: Event where you know one of the room not containing the Mercedes

Prove that $\mathbb{P}(\mathcal{E}_1) < \mathbb{P}(\mathcal{E}_2 \mid \mathcal{C})$

20. A group of n men enter a restaurant and check their hats. The hat-checker is absent minded, and upon leaving, she redistributes the hats back to the men at random. What is the probability P_n that no man gets his correct hat, and how does P_n behave as n approaches infinity?

21. You are given that at least one of the events A_n, $1 \leq n \leq N$, will occur almost surely i.e. $\mathbb{P}(\cup_{n=1}^{N} A_n) = 1$. Also no more than two occur almost surely. If $\mathbb{P}(A_n) = p$ and $\mathbb{P}(A_n \cap A_m) = q$, $m \neq n$, then show that $p \geq \frac{1}{N}$ and $q \leq \frac{2}{N}$.

22. Three friends are hanging out. To keep things simple, assume their birthdays are uniformly and independently distributed across the 365-day year. What's the probability that some (i.e. either two or three) of the people in this group have a birthday in common?

23. You invite some friends to a hat-themed party. As the night progresses, you decide to play a game: everyone will line up and place their hat on the ground in front of them. The guests will then be reordered randomly, with all orderings equally likely. Afterward, each guest will grab the hat in front of them.

    We're interested in the probability that at least one guest is wearing their own hat at the end of the game. What does that probability converge to as the number of guests grows infinitely large?

    Let A_i be the event that i^{th} person picks own hat. For example A_5 represents that 5th person picks own hat.

    We are interested in probability that at least one guest is wearing their own hat which is $\mathbb{P}(\bigcup_{i=1}^{N} A_i)$.

24. Let A, B be events such that $\mathbb{P}(A) = \frac{1}{4}$ and $\mathbb{P}(B) = \frac{1}{2}$. Find maximum possible values for $\mathbb{P}(A \cup B)$ & $\mathbb{P}(A \cap B)$

25. Assume a certain city X experience a hot day day with probability 0.4 and a cold day with probability 0.6. Each hot day experiences rain with probability 0.2 while a cold day experiences rain with probability 0.7. Given a particular day is rainy, what is the probability that the day is hot.

26. Let A_i be a sequence of events such that $P(A_i) = \frac{1}{2^i} \ \forall i \in \mathbb{N}$.$A_i$'s partition the sample space. Let C be another event such that $P(C|A_i) = \frac{1}{3^i} \ \forall i \in \mathbb{N}$. Find $P(C)$

27. Let A and B be events such that $\mathbb{P}(A|B) > \mathbb{P}(A)$. Show that $\mathbb{P}(B|A) > \mathbb{P}(B)$ and $\mathbb{P}(A|B^c) < \mathbb{P}(A)$.

28. A positive integer is selected, with $P(I = n) = \frac{1}{2^n}, n = 1, 2, \cdots$. If I takes the value n, a coin with probability of heads equal to e^{-n} is tossed once. Find the probability that the resulting toss will be a head.

29. Given a sequence of events $A_1, A_2, \cdots, A_n$ satisfying

$$\sum_{i=1}^{n} \mathbb{P}(A_i) = \mathbb{P}(\bigcup_{i=1}^{n} A_i).$$

Prove that $\mathbb{P}(A_i \cap A_j) = 0 \; \forall \; i, j, i \neq j$.

30. A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.